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Paleomagnetic evidence for counter-clockwise rotations in the southern Apennines fold-and-thrust belt during the Late Pliocene and middle Pleistocene

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Abstract

Marine and continental (clay/silt) sediments from Late Miocene to middle–Late Pleistocene age have been sampled in the southern Apennines fold-and-thrust belt to detect the spatial and temporal aspects of tectonic rotations. We sampled different parts of the tectonic system: sites from both the internal (western) and the external (eastern) part. The characteristic remanent magnetisation (ChRM) component for the sediments is generally removed from 320–340° up to 450–480°C. A counter-clockwise rotation is observed for nearly all sites. All data indicate that the main tectonic rotations in the southern Apennines fold-and-thrust belt occurred in the middle Pleistocene. The Late Miocene to middle Pliocene sediments reveal a 39° counter-clockwise rotation, which is similar to the rotation of the Cretaceous carbonate rocks. This indicates at least two phases of counter-clockwise rotation: a first rotation through approximately 15° during the Late Pliocene, possibly contemporaneous with clockwise rotations in Sicily (3.05 Ma), and a second rotation through at least 15° during the middle Pleistocene; there was also a third, smaller rotation through about 9° since the middle Pleistocene. The results imply that there were no tectonic rotations in the southern Apennines fold-and-thrust belt during the period of major shortening that took place in the Early–Middle Miocene, nor during the opening of the southern part of the Tyrrhenian basin in the latest Pliocene–earliest Pleistocene. Our data further support the hypothesis that the late oroclinal rotations in the Tyrrhenian arc were confined to distinct phases of regional compression.

1. Introduction

The geodynamical evolution of the southern Italian area since the Late Pliocene is dominated by extension related to the migration of the Calabrian arc to the southeast and the opening of the Tyrrhenian back-arc basin (Fig. 1). The extension is characterised by a radial distribution of stress (Philip, 1987), connected with concentric normal fault systems within the Tyrrhenian basin, and NW–SE-trending transtensional faulting within the Calabro-Peloritan block (Van Dijk and Okkes, 1991). In the southern Apennines fold-and-thrust belt, tectonic activity and sedimentation shifted eastwards and southwards during the Neogene and Quaternary. Passive subduction of a

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lithosphere slab by gravitational sinking is generally regarded as the geodynamic process responsible for this evolution (Ritsema, 1972; Malinverno and Ryan, 1986). Some authors concluded that major block rotations were probably induced by this arc migration process (Knott and Turco, 1991). From earlier studies, however, we deduced that block rotations did not occur simultaneously with the major phase of extension in the Tyrrhenian basin (Scheepers et al., 1993, 1994).

The area between Napoli and Bari (southern Italy) can be divided in three main tectonic units from more internal (west) to more external (east). These zone are: the southern Apennines Fold Belt, the Bradano Unit and the Apulia–Gargano foreland. The most easterly and external one, the Apulia–Gargano foreland (Fig. 1), consists of a Palaeozoic basement covered by a carbonate succession of Mesozoic–early Tertiary age that exceeds 6 km in thickness. Its Late Pliocene–Pleistocene marine clay cover was studied earlier and revealed no tectonic rotations (Scheepers, 1992).

The Apulian Unit extends to the west, structurally below the deposits of the Bradano Unit (Fig. 1), a succession of marine clays that was accumulated in a Plio-Pleistocene foredeep. These young foredeep sediments of the eastern part of the Bradano Unit and the underlying Mesozoic–early Tertiary carbonate sequence are affected by some minor normal faulting (Cristofolini et al., 1985). Compressive structures are detectable in the western part of the Bradano Unit, and are related to the thrusting of the

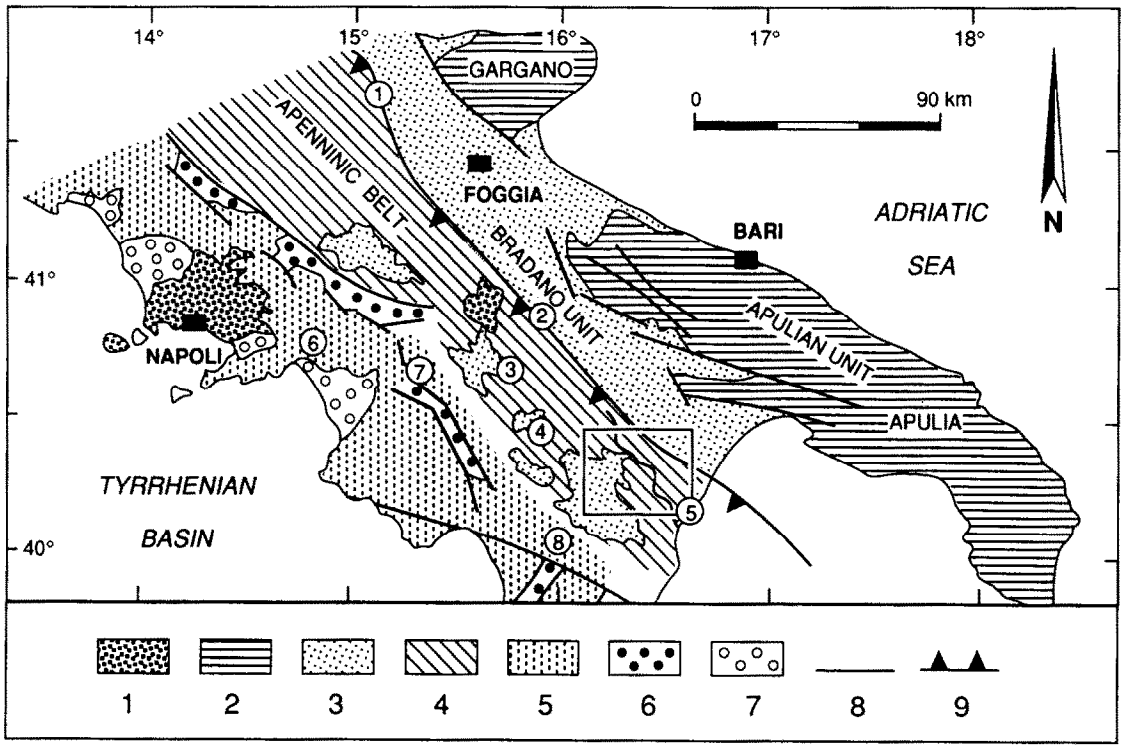


Fig. 1. Sampled areas, and major tectonic units in the southern Apennines system (modified after Albarello et al., 1990; Cristofolini et al., 1985). Legend: *1* = volcanic rocks; *2* = Apulia–Gargano foreland; *3* = Pliocene–Pleistocene deposits; *4* = external Apenninic belt; *5* = internal Apenninic belt; *6* = internal grabens; *7* = internal "Peri-Tyrrhenian" deposits; *8* = faults; *9* = buried Apenninic thrust front. Sampling localities: *(1)* = Casalnuovo Monterotaro (CNM); *(2)* = Spinazzola (SPZ); *(3)* = Potenza (POT); *(4)* = Calvello (CAL); *(5)* = Matera area (studied earlier; Scheepers et al., 1993); *(6)* = Salerno (SLN); *(7)* = Éboli (EBO); *(8)* = Rotonda (ROT).

southern Apennines Fold Belt probably during the middle Pleistocene (Casnedi et al., 1982). The Late Pliocene–Pleistocene deposits of the Bradano Unit were studied earlier and show no tectonic rotation (Scheepers et al., 1993). The eastern part of the southern Apennines Fold Belt (Fig. 1) is characterised by compressive structures of Plio-Pleistocene age (Cristofolini et al., 1985). Our previous paleomagnetic study of Early Pleistocene clays from the Matera region showed a counter-clockwise rotation over approximately 23° for the southeastern part of the southern Apennines Fold Belt (Fig. 1; Scheepers et al., 1993). This rotation is thought to have occurred during the middle Pleistocene, between 1.0 and 0.7 Ma, and is most probably connected to a middle Pleistocene compressive phase. We also found tectonic rotations in the opposite sense (15° clockwise) during the middle Pleistocene for the Calabro-Peloritan block (Fig. 1; Scheepers et al., 1994) which is situated to the south of the southern Apennines fold-and-thrust belt (Fig. 1). Paleomagnetic data of Cretaceous rocks from the southern Apenninic belt also imply counter-clockwise rotations (Channell and Tarling, 1975; Catalano et al., 1976; Incoronato, 1983; Jackson, 1990). Paleomagnetic results from the sites in the internal part (more western) of the southern Apenninic belt show a larger counter-clockwise rotation (89°) than the sites from the more external (eastern) part. Studies in Mesozoic rocks from three areas in this (external) part of the southern Apennines Fold Belt show an average of 37° counter-clockwise rotation (Table 4). The western (Tyrrhenian) part of the southern Apennines Fold Belt was affected by extension since the Late Miocene (Cristofolini et al., 1985; Albarello et al., 1990). This extension continued during the Quaternary, whereas the rest of the Tyrrhenian arc underwent intense uplift over up to 1000–1200 m (Bousquet, 1972). This study details results from additional sites in the southern Apennines Fold Belt in order to test for lateral consistency of the counter-clockwise rotations observed from previous studies in the Matera area (Fig. 1), and to further study the various counter-clockwise rotations observed for the Cretaceous (37°) and the Early Pleistocene (23°) rocks.

Sites studied here are spread across the southern Apennines Fold Belt, both in its internal and external parts (Fig. 1), and are additional to the earlier studied sites in the Matera region located further to the southeast.

2. Sampling and laboratory treatment

Sampling sites were selected mainly on the basis of sedimentary and lithological criteria. Fine-grained sediments were chosen, on the grounds of their suitable magnetic content and because their sedimentation rate is usually low enough to average out secular variations affects at site level. Sections and/or outcrops close to major faults or containing landslides were avoided. In some cases it was necessary to remove the weathered surface of the outcrop to a depth of 50 cm. As a rule, we drilled a minimum of six cores of 25 mm diameter and 6–10 cm length using a water-cooled electrical drill and portable generator.

The present study deals with Tertiary and Quaternary marine clays/marls from the southern Apennines Fold Belt and from the Bradano Unit. A total of 150 cores was sampled from 22 sites in seven different localities (Fig. 1), north and west of the earlier studied Matera area (Scheepers et al., 1993). In the external part of the southern Apennines Fold Belt (and the Bradano Unit), four localities were sampled in sediments unconformably overlying major thrust sheets. The sites are located from north to south (Fig. 1): in the Casalnuovo Monterotaro area (1), in the Spinazzola area (2), in the Potenza basin (3) and in the Calvello area (4). A further three localities were sampled in the internal (Tyrrhenian) part of the southern Apennines Fold Belt (Fig. 1): in the Salerno area (6), in the Éboli area (7) and in the Rotonda area (8). The Matera area (5) has been studied earlier by Scheepers et al. (1993).

2.1. The external part of the southern Apennines

(1) West of the Gargano peninsula, we drilled three sites (CNM 1–3) in optimally fresh horizon-

tally bedded clays of early Pleistocene age in an active quarry east of the village of Casalnuovo Monterotaro (Fig. 1: loc. 1).

(2) In the Bradano Unit, just west of the Apulian Unit, we furthermore sampled an (inactive) quarry north of the village of Spinazzola (Fig. 1: loc. 2), where we drilled three sites (SPZ 1–3). In these sites, bedding tilt is only moderate (7°) to the east-northeast. The cores were taken from blue-coloured clays.

(3) In the Potenza basin (Fig. 1: loc. 3), we sampled three sites (POT 1–3) in middle Pliocene clays. Sites POT 1–2 are in two different quarries north of the city of Potenza where bedding tilt is only slight (4–8°). Two kilometres west of Potenza, just north of the motorway, there is an outcrop in coarse conglomerates which show tilting of 29° to the north. Here, we sampled some intercalated, small-scale (20 m) grey fluviatile clay lenses (site POT 3).

(4) Further to the south, we sampled in the Calvello area (Fig. 1: loc. 4); where we drilled four sites in silts/clays of presumably middle Pliocene age (F.J. Hilgen, pers. commun., 1993). Sites CAL 1–2 are situated in a very fresh outcrop in grey silty clays just east of the village Calvello, which show bedding tilts of 20° to the north-northeast. One kilometre further to the east in the "Fosso la Terra" valley we sampled site CAL 3 in blue-grey siltstones, comprising calcareous fragments, where bedding tilt is 21° to the northeast. Site CAL 4, about 1 km east of the previous site, was sampled in grey-blue-coloured silty clays; they show a bedding tilt of 23° to the north.

2.2. *The internal part of the southern Apennines*

In the internal part of the southern Apennines we sampled at three different localities: the Salerno area, the Éboli area and the Rotonda area. The most internal one is situated in between the villages of Salerno and Éboli (Fig. 1: locs. 6 and 7).

(6) Near Salerno, we drilled four sites: south of Montecorvino Rovella, we sampled one site (SLN 1) in a quarry, which did not permit any bedding tilt control. Three sites (SLN 2-4) were sampled in clays of (most probably) late Tortonian–Early Pliocene age (F.J. Hilgen, pers. commun., 1993), directly north of Salerno along motorway A3. These grey-blue clays are badly exposed and contain some sandy intercalations which show a bedding tilt of 16.5° to the south. The sites cover stratigraphically some 20 m.

(7) In the Éboli area, we sampled seven sites. Directly to the west of Éboli, in an active quarry, we sampled lagoonal marly sediments (EBO 7). Some 8 km east of Éboli, Early Pliocene clays occur, dipping 32° to the southwest, covered by (sub)horizontal Pleistocene sands. The clays are dark-grey and compact, and contain some intercalated sandy layers. The three sampled sites (EBO 4–6) cover 25 m stratigraphically. Further to the east, we sampled three sites near Sicignano, two of which (EBO 1–2) are in blue clays on top of Pleistocene conglomerates. The age of the clays is Early–middle Pleistocene (A. Cinque, pers. commun., 1992) and they show a bedding tilt of 7° to the south-southwest. Site EBO 3 is taken in lacustrine limestones of Late Pleistocene age.

(8) In the most southern part of the southern Apennines, 20 km west-northwest of the Monte Pollino near the villages Rotonda and Castellúccio (Fig. 1: loc. 8), we sampled some outcrops in continental sediments of (middle–Late) Pleistocene age. Site ROT 1 is situated in fluviatile sediments, 2 km north of Rotonda in the valley of the Mércure river. An outcrop in lagoonal sediments 1 km southeast of Castellúccio Inferiore was sampled (ROT 2–3) as well; it consists of grey-brown clays with calcareous fragments. The age of these rocks corresponds to the Mindel–Würm interglacial (0.3–0.1 Ma) (Bousquet and Gueremy, 1968). Two kilometres west of Castellúccio Inferiore, we sampled an outcrop in lagoonal marls (ROT 4). The orientation of the bedding plane is (sub)horizontal for all these outcrops in the southernmost area.

Routine paleomagnetic methods were applied in the laboratory in order to determine the characteristic magnetisation directions. The natural remanent magnetisation (NRM) was measured partly with a JR-3 spinner magnetometer with a digital interface, and partly with a 2G cryogenic

magnetometer. The NRM of generally six specimens per site was analysed by means of progressive stepwise thermal demagnetisation using small temperature increments (50–20°C). Changes in the low-field bulk susceptibility (X_{LF}) of the samples during thermal treatment were monitored with a Bartington MS2 susceptometer, in order to detect changes in the magnetic mineralogy. The final part of the thermal demagnetisation was limited by a large increase in X_{LF} and viscous behaviour after heating at temperatures above 400–450°C. Increasingly viscous behaviour and low remanent intensities impeded determination of meaningful remanence directions. Demagnetisation diagrams (Zijderveld, 1967) were used for the determination and interpretation of characteristic remanence magnetisation directions. The characteristic directions of the magnetisations were determined by least squares fitting (principal component analysis) through selected datapoints (Kirschvink, 1980). From these ChRM directions mean directions were calculated for each site according to Fisher (1953).

3. Results

3.1. The external part of the southern Apennines

Previously obtained paleomagnetic results from the southern Apennines Fold Belt of the Matera area showed significant counter-clockwise rotations for Early Pleistocene clays (Scheepers et al., 1993): three localities give an average counter-clockwise rotation of 23° ($k = 254$). Previously studied sampled clays of the Bradano Unit in the eastern part of the Matera area, however, show no tectonic rotation (Scheepers et al., 1993). Pale-

Table 1
Site mean directions from the sites of this study using Fisher (1953) statistics

Area	Site	N	D	I	D(tc)	I(tc)	k	α_{95}	Int	Temp
External part of the southern Apennines										
(1)	CNM 1	6	145.9	−67.6	145.9	−67.6	308	3.8	3.09	HT
	CNM 2	6	156.6	−66.4	156.6	−66.4	161	5.3	5.24	HT
	CNM 3	7	159.9	−62.0	159.9	−62.0	263	3.7	3.37	HT
(2)	SPZ 1	4	188.3	−53.3	178.9	−55.3	155	7.4	0.69	HT
	SPZ 2	5	185.7	−60.1	173.4	−61.6	197	5.5	0.89	HT
	SPZ 3	7	191.7	−52.5	182.7	−54.9	221	4.1	0.34	HT
(3)	POT 1	6	166.5	−45.6	162.5	−44.9	8	25.7	0.06	HT
	POT 2	4	168.8	−36.6	164.8	−42.9	11	29.1	0.06	HT
	POT 3	8	93.2	−79.1	171.4	−61.2	10	18.2	0.04	HT
(4)	CAL 1	6	323.9	38.4	340.9	42.5	45	10.1	0.07	LT
	CAL 2	5	333.4	46.4	354.7	46.4	195	5.5	0.09	LT
	CAL 1	6	274.7	38.6	284.2	56.8	104	6.6	0.23	HT
	CAL 2	5	275.7	39.1	285.9	57.1	293	4.5	0.27	HT
	CAL 3	6	302.0	48.1	327.1	51.7	781	2.4	0.65	HT
	CAL 4	6	302.3	67.0	328.8	50.4	191	4.9	0.93	HT
Internal part of the southern Apennines										
(6)	SLN 1	6	330.4	−66.3	149.7	−53.7	41	10.6	0.11	HT
	SLN 2	4	329.4	40.8	322.7	56.2	73	10.8	0.11	HT
(7)	EBO 1	5	357.7	56.3	350.0	61.6	107	7.4	0.16	HT
	EBO 2	6	6.6	53.2	1.2	59.2	122	6.1	0.65	HT
	EBO 4	6	340.5	38.3	313.4	43.9	45	10.1	0.59	HT
	EBO 6	6	345.3	40.8	312.2	48.0	104	6.6	1.38	HT
(8)	ROT 2	6	346.9	48.5	346.9	48.5	316	3.8	1.97	HT
	ROT 3	6	348.8	54.7	348.8	54.7	1050	2.1	4.74	HT

N = number of specimens out the 6–8 cores taken per site; D(tc), I(tc) = declination, inclination (corrected for bedding tilt); k = precision parameter; α_{95} = semi-angle of cone of confidence on the 95% level; Int = mean intensity per site (in mA/m); temp = temperature.

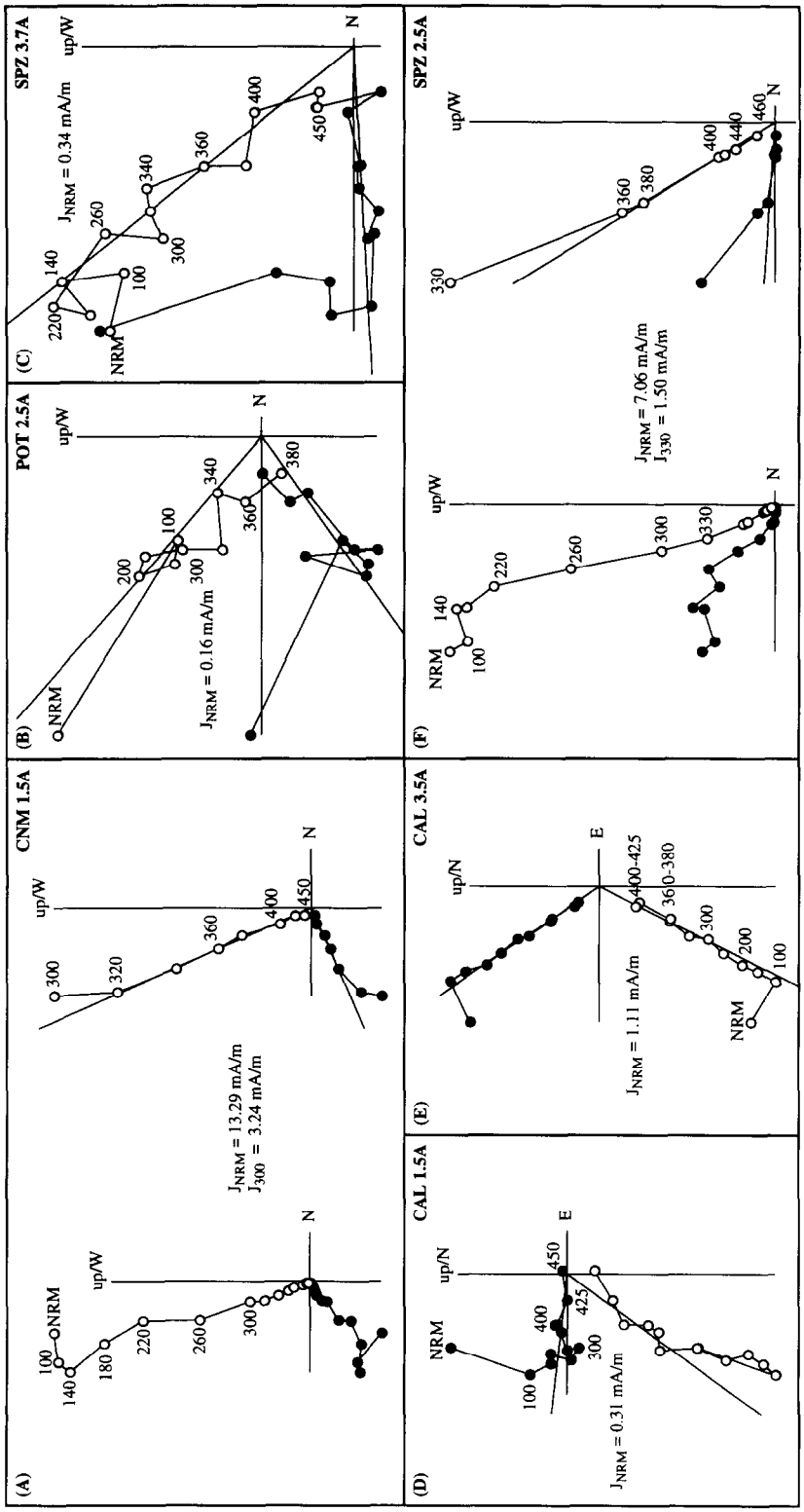


Fig. 2. Stepwise thermal demagnetisation diagrams (corrected for bedding tilt) of representative specimens from the external part of the southern Apenninic belt system. Numbers denote demagnetisation steps in °C. Direction and intensity of magnetisation during thermal demagnetisation were analysed from orthogonal projection plots (Zijderveld, 1967) and principal component analysis (Kirschvink, 1980). ●indicate projections on the horizontal plane, ○ indicate projections on a vertical plane. Abbreviations: *CNM* = CasaInuovo Monterotaro; *POT* = Potenza; *SPZ* = Spinazzola; *CAL* = Calvello.

Table 2
Mean site directions for the localities of this study before (no tc) and after (tc) bedding tilt correction

Area	N	Tc	D(tc)	I(tc)	k	α_{95}
Casalnuovo Monterotaro (CNM 1–3)						
(1)	**3 sites**	**no tc**	**154.6**	**−65.4**	**370**	**6.4**
Spinazzola (SPZ 1–3)						
(2)	**3 sites**	no tc	188.8	−55.3	317	6.8
		tc	**178.6**	**−57.3**	**317**	**6.8**
Potenza (POT 1–3)						
(3)	**3 sites**	no tc	161.0	−55.7	9	44.1
		tc	**165.6**	**−49.7**	**61**	**16.0**
Calvello (CAL 1–4)						
(4)	**4 sites**	no tc	286.0	48.9	24	19.2
		tc	308.1	55.9	30	16.9
		no tc (c)	305.0	48.2	37	15.3
		tc (c)	**322.8**	**54.1**	**247**	**5.9**
Eboli (EBO 1–2						
(5)	**11 samples**	no tc	2.7	54.7	107	4.4
		tc	**356.3**	**60.4**	**107**	**4.4**
Salerno – Eboli (SLN 1–2 / EBO 4–6)						
(5–6)	**4 sites**	no tc	340.1	56.4	5	45.2
		tc	**318.9**	**50.7**	**113**	**8.7**
Rotonda (ROT 2–3)						
(7)	**12 samples**	**no tc**	**347.8**	**51.6**	**284**	**2.6**

(c) refers to a correction for sites CAL 1–2 for the deviation of the LT component (see text). The preferred result is shown in bold. Other symbols are the same as in Table 1.

omagnetic results for the sites studied here are very similar to the results from the Matera area (Fig. 1). A stable high-temperature (HT) component is removed between 320–340 and 450°C. The three localities in the external part of the southern Apennines Fold Belt situated north of the Matera area give counter-clockwise rotations for the HT component.

(1) The Casalnuovo Monterotaro area

The sites in Early Pleistocene clays near Casalnuovo Monterotaro show very high initial NRM intensities (12–70 mA/m). The direction of the HT component is well determinable for the three sites (Table 1; Fig. 2A) and shows significant counter-clockwise rotations. The mean direction for the three sites is $D_m = 154.6°$ and $I_m = -65.4°$ ($k = 370$; Table 2), implying approximately 25° of counter-clockwise rotation for these deposits. This is similar to the rotation established previously for the Early Pleistocene clays of the Matera area, some 200 km to the southeast.

(2) The Spinazzola area

The three sites sampled near Spinazzola in Early Pleistocene clays from the Bradano Unit have initial NRM intensities between 2.2 and 17.2 mA/m, although cores from some brownish (weathered) clays (Fig. 2C) have considerably lower NRM intensities around 0.3 mA/m. The paleomagnetic results are very similar to the results in Early Pleistocene clays from the Casalnuovo Monterotaro and Matera areas (Fig. 2A); a high-temperature (HT) component is removed between 320–340 and 450°C. The low-intensity sample SPZ 3.7 from the brown clays (Fig. 2C) gives a direction for the HT component that is similar to the high-intensity cores from the blue clays (Fig. 2F). This indicates the good stability of the HT component in both the low- and high-intensity samples. The direction of the HT component is very well determinable for the three sites (Table 1) and shows no rotation. The mean direction for the three sites in Early Pleistocene clays is $D_m = 178.6°$ and $I_m = -57.3°$ ($k = 317$; Table 2), implying no rotation for the Spinazzola clays.

(3) The Potenza area

The three sites from the Potenza basin show very low initial NRM intensities (below 0.2 mA/m). Although individual demagnetisation diagrams show quite scattered trajectories (Fig. 2B), we calculated a magnetisation direction for each sample using the temperature steps between 250 and 350°C. From 360–380°C upwards the amount of (viscous) scatter relative to the NRM (then below 0.01 mA/m) becomes to large to measure any meaningful directions (Fig. 2B). The mean directions for the three sites show low values for the precision parameter (Table 1). The significant bedding tilt of site POT 3 allows a meaningful fold test. The mean direction for the three sites, $D_m = 165.6°$ and $I_m = -49.7°$ ($k = 61$; Table 2), shows a significant improvement of clustering after correction for bedding tilt (Table 2), implies approximately 14° of counter-clockwise rotation for the Potenza basin.

(4) The Calvello area

The sites in silts and clays of middle Pliocene age from the Calvello area show a simple behaviour upon demagnetisation and well determinable directions (Fig. 2D, E). NRM intensities range between 0.3 and 2.2 mA/m. For sites CAL 1–2 (Fig. 2D) a low-intensity (70–90 mA/m) normal polarity component is removed between 100 and 200°C. The direction of this low-temperature component (LT) shows a 32° counter-clockwise deviation from the north (in-situ) (Table 1; Fig. 3C). The main (characteristic) magnetisation component was removed between 300 and 450°C; this trajectory was used to calculate the HT component (Table 1). There is a considerable difference in rotation of the HT component between on the one hand sites CAL 1–2 (near the village), and on the other hand sites CAL 3–4 which were sampled more to the east (Table 1; Fig. 3B). The amount of rotation observed for the easterly sites is comparable to the rotation determined for the Matera, Potenza and Casalnuovo Monterotaro areas. The additional 40° of counter-clockwise rotation for sites CAL 1–2 is not well understood, but probably reflects a very local and recent phenomenon. This follows from the 32° counter-clockwise rotation (in-situ) of the LT component (Fig. 3C). If we correct the HT direction of sites CAL 1–2 for this 32° counter-clockwise deviation, assuming no rotation for the LT component (Dec = 0°), there is a remarkable similarity in direction of the HT components with sites CAL 3–4 (Fig. 3D). The mean direction for the four sites in the quarry is then $D_m = 322.8°$ and $I_m = 54.1°$ ($k = 247$; Table 2), still implying 37° of counter-clockwise rotation for the clays from the Calvello area.

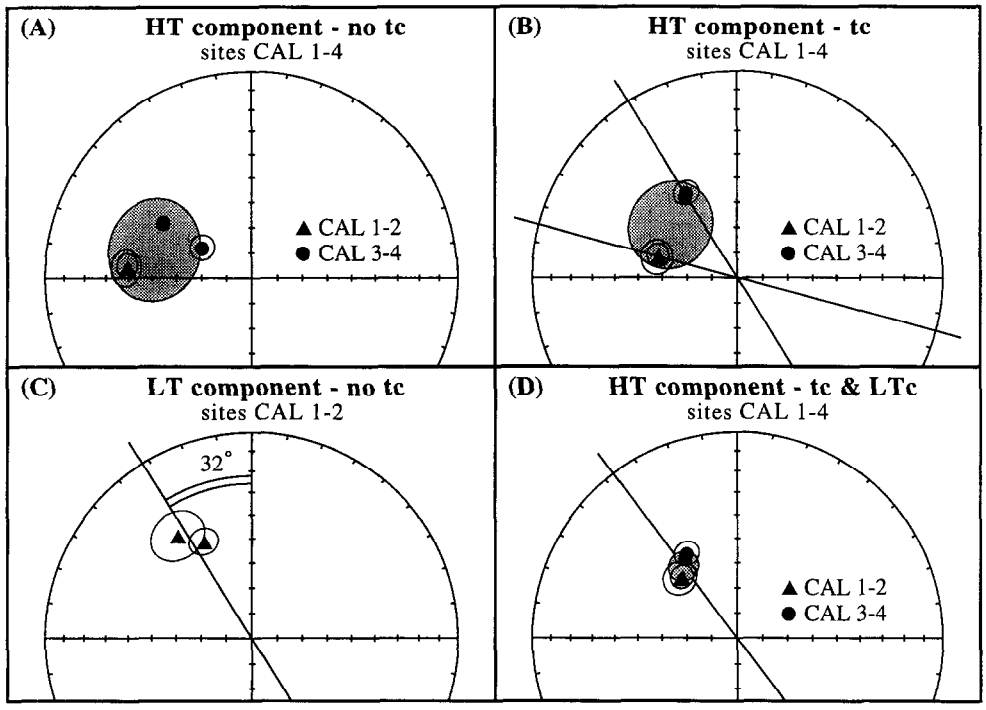


Fig. 3. Equal-area stereographic projections of high-temperature (HT) magnetisation site mean directions of the Calvello area (*CAL*). For all diagrams, circles give Fisher's (1953) α_{95} for the different site means; shaded area gives the overall mean direction for the area using the site mean directions. (A) In-situ high-temperature (HT) directions. (B) Tilt-corrected HT directions. (C) In-situ low-temperature (LT) directions show a 32° counter-clockwise deviation. (D) Tilt-corrected HT directions; sites *CAL 1–2* were additionally corrected for the 32° counter-clockwise deviation of the LT direction (LTc) from diagram (C).

3.2. The internal part of the southern Apennines

(6) The Salerno area

Only two sites (SLN 1–2) from the four that we sampled in Late Miocene clays near the village of Salerno give good paleomagnetic results. Because we have no bedding control for site SLN 1, we assumed that the magnetic foliation determined from the anisotropy of magnetic susceptibility is parallel to the bedding plane (see also Scheepers and Langereis, 1994). If we apply this correction to the site, the remanence magnetisation direction has a reversed polarity and a counter-clockwise rotation of 30° (Table 1; Fig. 4A). The other three sites (SLN 2–4), taken from the section north of Salerno, provided a clear bedding plane control. Only SLN 2 shows a reliably determinable remanence direction (Fig. 4B). A normal polarity component is removed up to 400°C (Fig. 4B), which shows a counter-clockwise rotation (37.5°) more or less comparable to that for site SLN1 (30.5°).

(7) The Éboli area

Only two of the sites in Early Pliocene clays near Éboli (EBO4 and EBO6) give good paleomagnetic results (Table 1; Fig. 4C). The polarity of the HT magnetisation removed up to 425–450°C is normal and the directions imply a significant counter-clockwise rotation (Tables 1, 2). These two sites show rotations comparable to those observed in the Salerno clays (Fig. 4C). The mean direction for the four sites in the Salerno–Éboli area is $D_m = 318.9°$ and $I_m = 50.7°$ ($k = 113$;

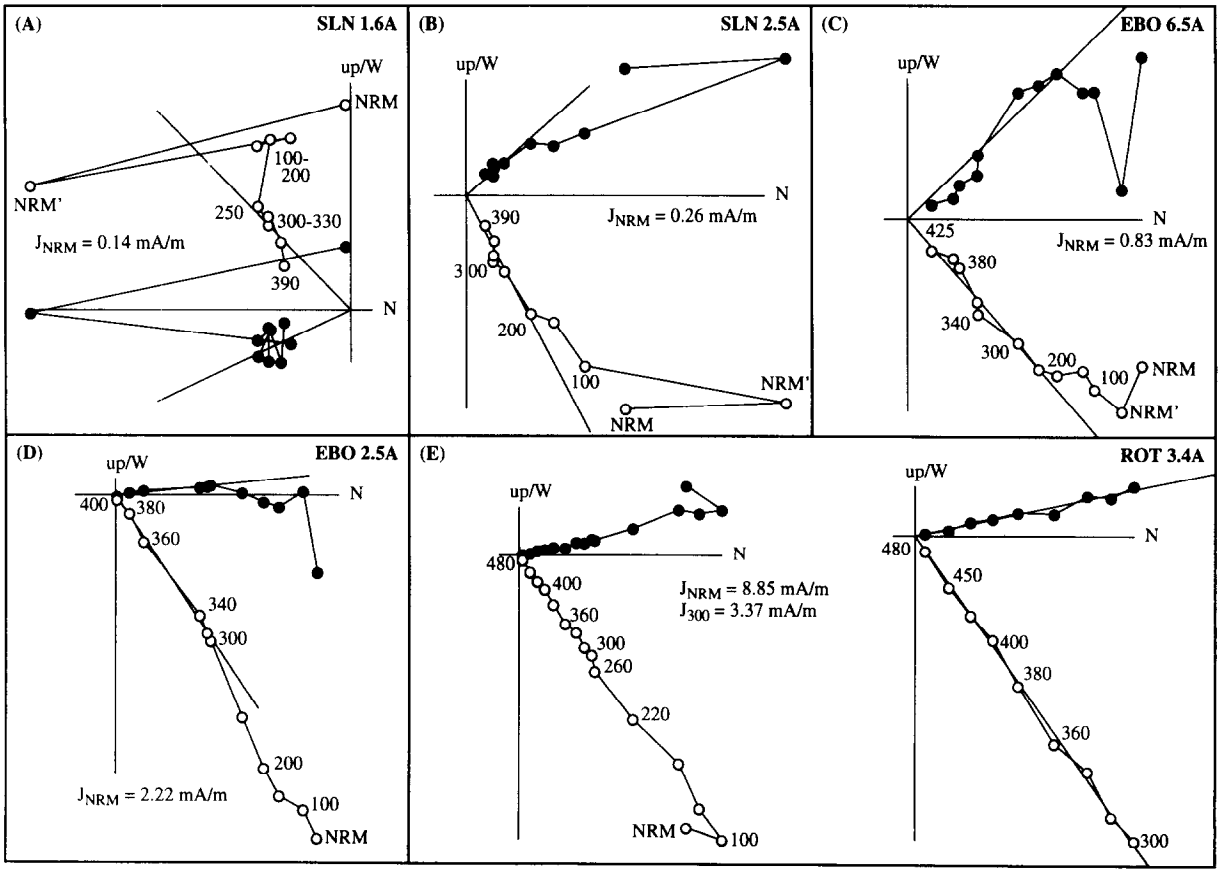


Fig. 4. Stepwise thermal demagnetisation diagrams (corrected for bedding tilt) of specimens from the internal part of the Apenninic belt system. Symbols are the same as used in Fig. 2. Abbreviations: *SLN* = Salerno; *EBO* = Éboli; *ROT* = Rotonda.

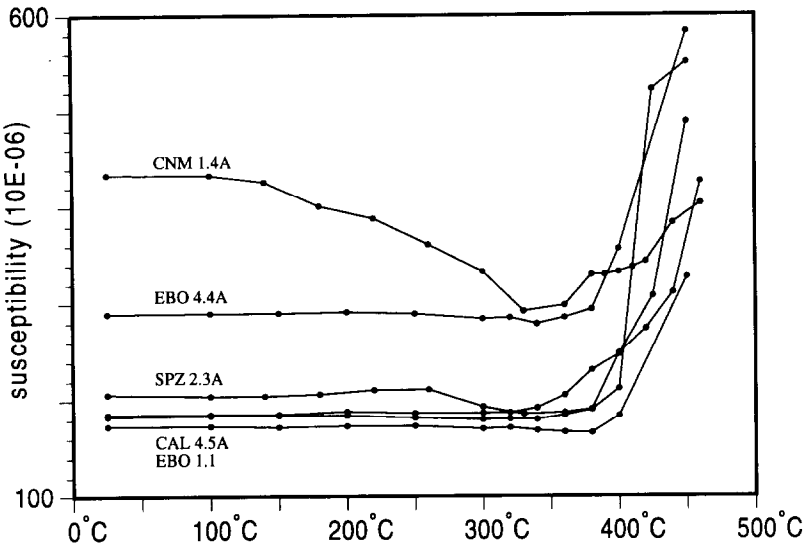


Fig. 5. Initial (bulk) susceptibility versus demagnetisation temperature for some specimens of this study. An increase in susceptibility at temperatures above 350–380°C is observed for all sites.

Table 2), implying on average 41° of counter-clockwise rotation.

The two sites in middle Pleistocene (non-marine) clays (near Sicignano, EBO 1–2) show a component which is removed up to 400°C (Fig. 4D). Its offset in declination is not very obvious. The mean direction for the eleven samples from these two clay sites, is $D_m = 356.3°$ and $I_m = 60.4°$ ($k = 107$; Table 2), implying no significant rotation. The other three sites for the Éboli area, EBO 3, 5 and 7, have extremely low NRM-intensities (below 0.05 mA/m) and directions are randomly oriented.

(8) The Rotonda area

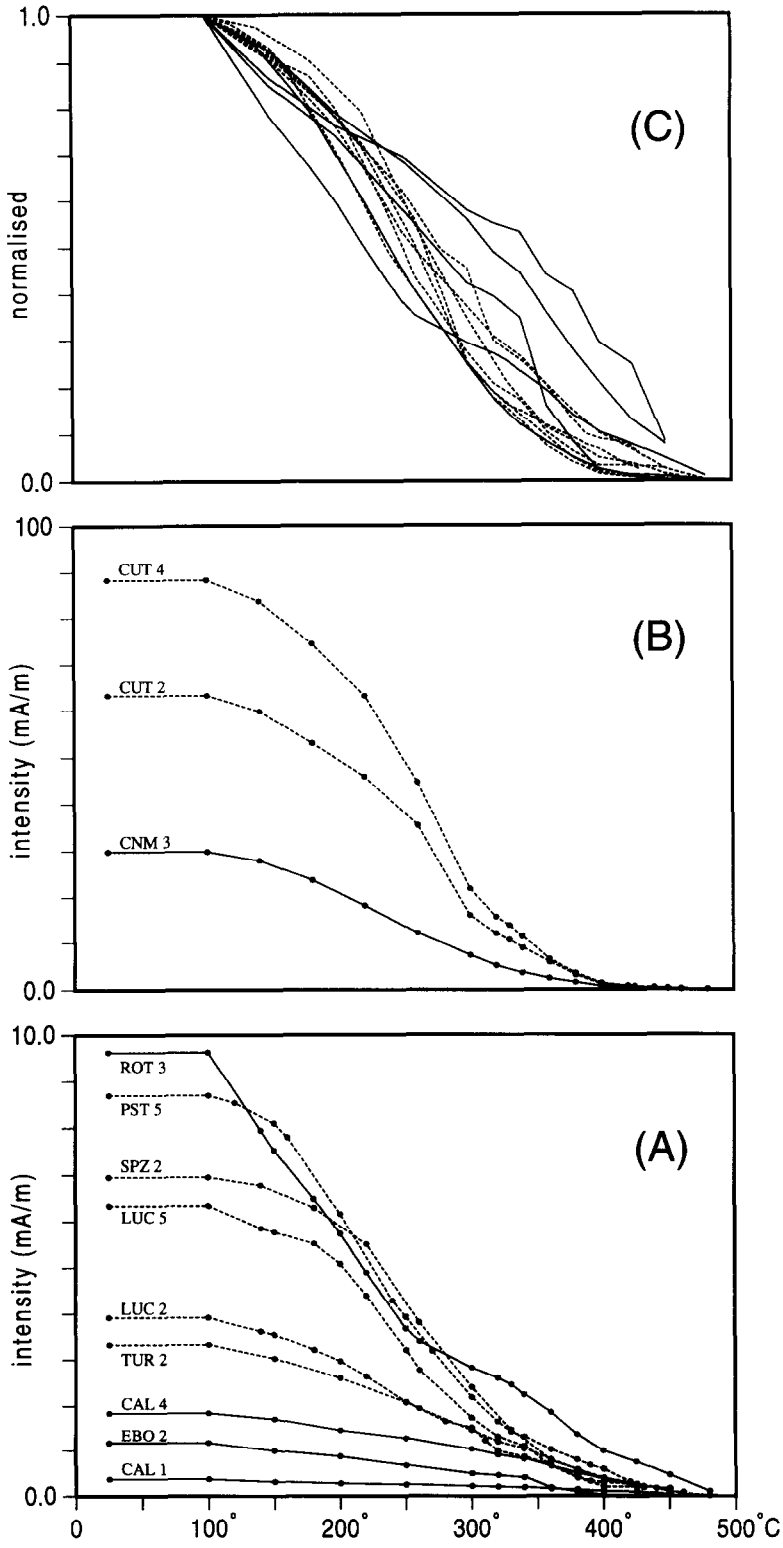
Only the outcrop directly southeast of Castellúccio Inferiore (sites ROT 2–3) give good paleomagnetic results (Fig. 4E). The initial NRM intensity for these middle–Late Pleistocene clays is 2.0–4.7 mA/m. In these sites, the magnetisation is also removed at 480°C. Both sites show normal polarity components that considering the young age of the sediments are taken to belong to the Brunhes Chron. The mean direction for the twelve demagnetised samples of sites ROT 2–3 is $D_m = 347.8°$ and $I_m = 51.6°$ ($k = 284$; Table 2), implying 12° of counter-clockwise rotation. Sites ROT 1 and ROT 4 have extremely low NRM-intensities (below 0.05 mA/m) and directions could not be determined precisely.

4. Discussion

Two components generally contribute to the NRM of the late Tertiary and Quaternary sediments from the southern Apennines fold-and-thrust belt. Most sites show a low-temperature (LT) component that is removed between 100 and 300–320°C (see also Scheepers et al., 1993)

→

Fig. 6. NRM decay curves (A, B) for representative specimens from 12 studied sites. (A) Intensity < 10 mA/m. (B) Intensity > 10 A/m. Solid lines give the sites from the southern Apennines Fold Belt; dotted lines the sites from the Bradano Unit and Apulia–Gargano foreland. Normalised curves (C) are quite uniform for all sites; maximum blocking temperatures are between 400 and 480°C.


1.0
normalised
0.0
(C)
100
intensity (mA/m)
CUT 4
CUT 2
CNM 3
0.0
(B)
10.0
intensity (mA/m)
ROT 3
PST 5
SPZ 2
LUC 5
LUC 2
TUR 2
CAL 4
EBO 2
CAL 1
0.0
(A)
0
100°
200°
300°
400°
500°C

and a high-temperature (HT) component that is removed between 300–320 and 430–450°C. In the latter temperature interval, the bulk susceptibility of the samples increases during thermal demagnetisation (Fig. 5), probably caused by formation of (close to superparamagnetic) magnetite

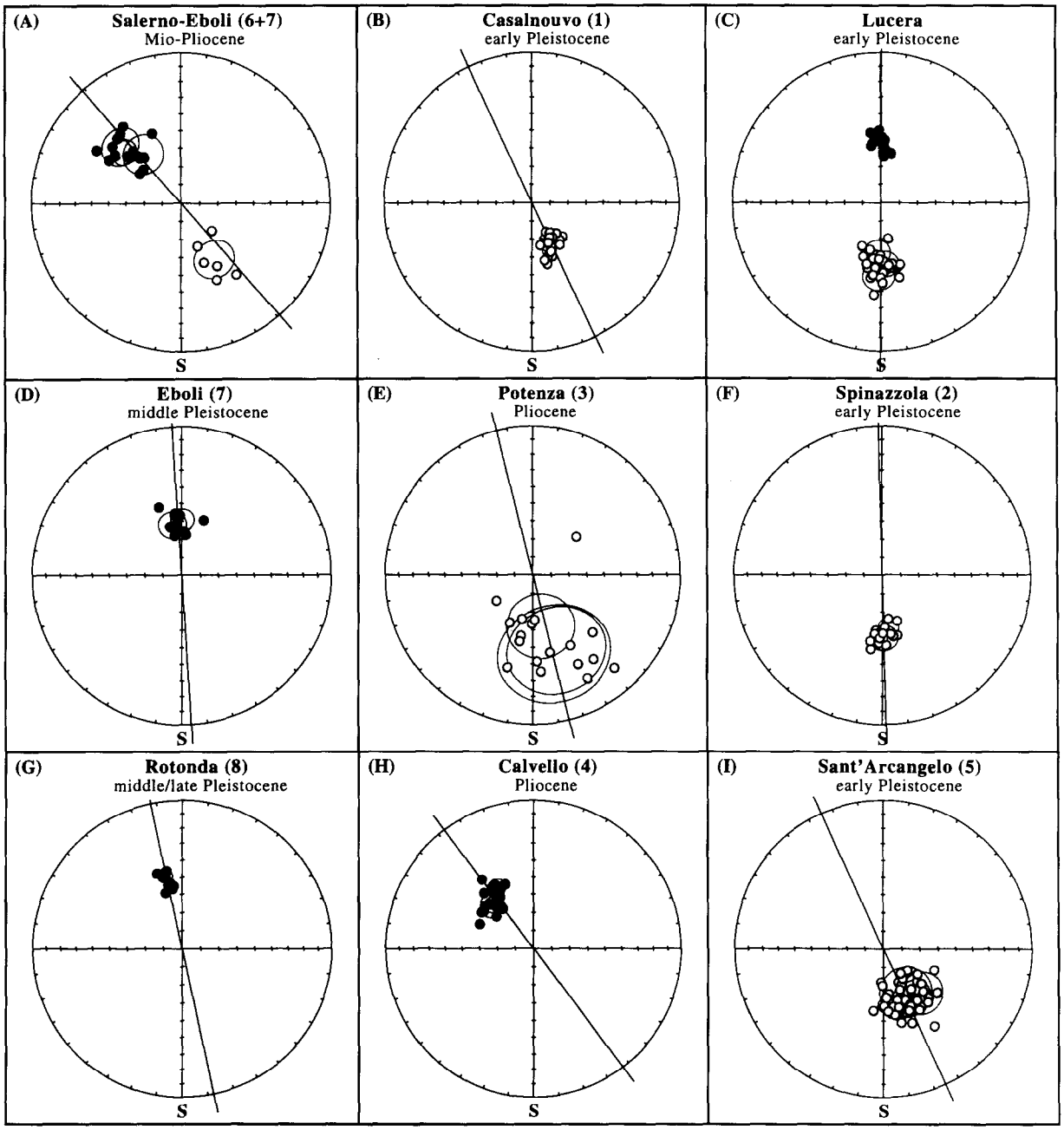


Fig. 7. Equal-area projection of HT components of specimens from the southern Apennines Fold Belt. •represent the individual sample directions; ○ denote downward (upward) directions. The circles give Fisher's (1953) α_{95} for the different site means. The lines represent the average tectonic rotations for the localities.

through oxidation of pyrite (see also Scheepers et al., 1993). The NRM decay-curves show a large variation in initial intensities (Fig. 6A, B). Normalisation of the curves, however, results in quite uniform shapes for the decay curves which points to a constant mineral assemblage carrying the NRM of these sediments even though they reflect a wide temporal and spatial frame (Late Miocene–middle Pleistocene, Tyrrhenian coast to Adriatic coast). The HT component may be carried by titanomagnetites (Scheepers et al., 1993). The HT magnetisation is affected by weathering resulting in a decrease in NRM intensity but this is not accompanied by a change in direction. Apparently, low-intensity samples can be used and (may) give reliable directions.

For reason of the consistent demagnetisation behaviour of the HT component in this interval, this component is taken to represent the characteristic remanent magnetisation direction for the clays and it is used as an indication of the tectonic rotation (Fig. 7). The studied sites show a

Table 3
Summary of main results from this paleomagnetic study

N	D(tc)	I(tc)	k(tc)	α_{95}	Pol	Locality	Age estimate	Ref
Southern Apennines Fold Belt								
6 sites	328.5	60.3	140	5.7	R	Sant'Arcangelo (SAG)-N	0.70–2.02 Ma	[II]
5 sites	343.0	57.2	122	7.0	R	Sant'Arcangelo (SAG)-S	0.70–2.02 Ma	[II]
6 sites	337.5	53.7	48	9.8	NR	Craco (CRC)	1.45–2.02 Ma	[II]
3 sites	334.6	65.4	370	6.4	R	Casaln. Monterotaro (CNM)	Early Pleist	[III]
3 sites	345.8	49.7	61	16.0	R	Potenza (POT)	middle Plio	[III]
4 sites	322.8	54.1	247	5.9	N	Calvello (CAL)	middle Plio	[III]
4 sites	318.9	50.7	113	8.7	NR	Salernoarea (SLN/EBO)	Late Mio–Plio	[III]
11 samples	356.3	60.4	107	4.4	N	Eboli (EBO)	< 0.5 Ma	[III]
12 samples	347.8	51.6	284	2.6	N	Rotonda (ROT)	< 0.5 Ma	[III]
4 sites	**351.3**	**56.1**	**144**	**7.7**	**N**	Apennines Fold Belt sites	< 0.5 Ma	
20 sites	**336.0**	**58.4**	**76**	**3.8**	**NR**	Apennines Fold Belt sites	Early Pleist	
8 sites	**320.8**	**52.4**	**159**	**4.4**	**NR**	Apennines Fold Belt sites	Late Mio–middle Plio	
Bradano Unit								
7 sites	359.4	56.6	98	6.1	R	Tursi section (TUR)	0.89–0.82 Ma	[II]
7 sites	4.2	51.6	88	6.5	NR	Pisticci section(PST)	1.05–0.90 Ma	[II]
3 sites	352.4	58.3	519	5.4	N	Pomaricoarea (POM)	< 0.99 Ma	[II]
3 sites	358.6	57.3	317	6.8	R	Spinazzola (SPZ)	Early Pleist	[III]
8 sites	0.4	55.9	299	3.2	NR	Lucera (LUC)	Early Pleist	[I]
3 sites	2.8	57.1	264	7.6	R	Ginosa (MSC)	Early Pleist	[I]
31 sites	0.6	55.6	136	2.2	NR	Bradano Unit sites		
10 sites	3.1	55.8	123	4.4	N	normal polarity sites		
21 sites	359.4	55.5	140	2.7	R	reverse polarity sites		
6 areas	**359.8**	**56.2**	**610**	**2.7**	**NR**	Bradano Unit localities		
Apulian Unit								
6 sites	2.2	61.2	46	10.0	R	Canosa (CAN)	Early Pleist	[I]
13 sites	354.3	52.9	81	4.6	R	Monte Mesola (MM)	Early Pleist	[I]
6 sites	0.2	59.1	1658	1.6	N	Cutrofiano (CUT)	middle Pleist	[I]
25 sites	358.6	57.8	278	1.7	NR	Apulian Unit sites		
6 sites	0.2	59.1	1658	1.6	N	normal polarity sites		
19 sites	356.4	55.6	58	4.4	R	reverse polarity sites		
3 areas	**358.6**	**57.8**	**278**	**7.4**	**NR**	Apulian Unit localities		

N = number of sites/samples; D(tc), I(tc) = declination, inclination (after bedding tilt correction); k(tc) = precision parameter (after tectonic correction); pol = polarity; ref = references, where [I] = Scheepers (1992), [II] = Scheepers et al. (1993) and [III] = Scheepers et al. (1994). Reverse polarity results are shown as their normal polarity equivalent to facilitate comparison with normal polarity data. The three sites of Potenza have not been used in the calculation of the Late Miocene–middle Pliocene mean direction because of their very low k's (Table 1).

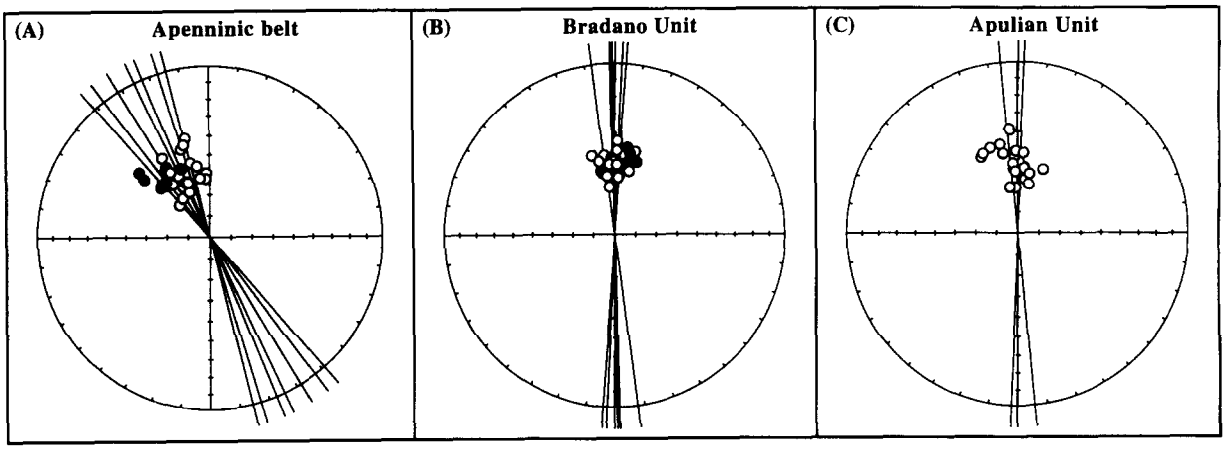


Fig. 8. Equal-area projection of site means from the southern Apenninic system. • represent the individual site directions; ○ denote normal (reverse) polarity sites. The lines represent the estimated tectonic rotations for the localities in the three parts of the system.

consistent pattern of paleomagnetic results (Fig. 7). A positive fold test for the HT component of the Potenza basin and the Salerno–Éboli area supports a primary (= pre-folding) origin. A positive fold test for the HT component was also observed for the sediments of the Sant'Arcangelo basin in the Matera area (Scheepers et al., 1993). The 31 sites (including Matera area) in the Apenninic belt system show without exception counter-clockwise rotations, ranging between 15 and 40° (Table 3; Fig. 8A). The sediments have ages between Late Miocene (±6 Ma) and Early Pleistocene (±0.7 Ma). The eight Late Miocene–middle Pliocene sites show the largest counter-clockwise rotation (Fig. 7A, H): 41° for the sites in the Salerno–Éboli area and 37° for the Calvello area with an average direction for the eight sites, $D_m = 320.8°$ and $I_m = 52.4°$ ($k = 159$; Table 3), that shows a 39° counter-clockwise rotation. The twenty sites in Early Pleistocene clays show, in contrast, over 24° counter-clockwise rotation ($D_m = 336.0°$ and $I_m = 58.4°$; $k = 76$; Table 3). Hence, we conclude to a middle Pliocene–Early Pleistocene counter-clockwise rotation over 15°, that may be related to the Pliocene rotational event for the Caltanissetta basin of Sicily (Scheepers and Langereis, 1993).

The four sites in middle–Late Pleistocene sediments from the internal part of the Apenninic belt (sites ROT 2–3 and EBO 1–2) show average directions of $D_m = 351.3°$ and $D_m = 56.1°$ ($k = 144$; Table 2). This implies a counter-clockwise rotation of approximately 9°.

The six localities in the Bradano Unit which include data presented previously by Scheepers (1992) from quarries near Lucera (LUC) and Ginosa (MSC) and by Scheepers et al. (1993) from sites near Tursi (TUR), Pisticci (PST) and Pomárico (POM), show an average direction of $D_m = 359.8°$ and $I_m = 56.2°$ ($k = 610$; Table 3). This implies that these sites (31 in total), which belong to the Early Pleistocene, do not show any tectonic rotation (Fig. 8B). From these 31 sites, the ten normal polarity sites give a mean direction of $D_m = 3.1°$ and $I_m = 55.8°$ ($k = 123$; Table 3), while the 21 reverse polarity sites give an overall mean direction of $D_m = 179.4°$ and $I_m = -55.5°$ ($k = 140$; Table 3). The angle between normal and reverse polarity mean directions is hence 2.1°, and the classification of the reversal test is "A" (McFadden and McElhinny, 1990).

Previously studied sites (25 in total) in Early–middle Pleistocene sediments from the Apulia–Gargano foreland (Scheepers, 1992) do not show any rotation either: $D_m = 358.6°$ and $I_m = 57.8°$ ($k = 278$; Table 3). The six normal polarity sites have a mean direction of $D_m = 0.2°$ and $I_m = 59.1°$ ($k = 1658$; Table 3), and the nineteen reverse

polarity sites show a mean direction of D_m = 176.4° and I_m = −55.6° (k = 58; Table 3). The angle between normal and reverse polarity mean directions is 4.1° and the reversal test classification is "A" (McFadden and McElhinny, 1990).

The HT component directions of the Early Pleistocene sediments from southern Italy do not show an inclination error: the average inclination is 58.4° for the Apenninic belt, 56.2° for the Bradano Unit and 57.8° for the Apulia–Gargano Unit (Table 3), which does not deviate significantly from the geocentric axial dipole inclination for the present latitude (57.5°).

The Early Pleistocene results indicate a 24° counter-clockwise rotation of the southern Apennines Fold Belt relative to the Bradano Unit and the Apulia–Gargano Unit from the foreland (Fig. 8). This rotation occurred largely during the middle Pleistocene. Results from middle to Late Pleistocene sites indicate a (possible) minor rotation over no more than 9° during the Late Pleistocene. This means that the rotation post-dates the major phase of extension that can be related to the formation of oceanic crust in the southeastern part of the Tyrrhenian basin, dated between 2.0 and 1.6 Ma (Sartori et al., 1989). The rotation may be related, however, to a subsequent compressive regime "the Quaternary compressive phase" that took place between 1.0 and 0.76 Ma (Philip and Tortorici, 1980; Auroux et al., 1986; Bousquet and Philip, 1986), as suggested before by Scheepers et al. (1993).

5. Conclusions

The mean direction for Cretaceous rocks from the Apulia–Gargano foreland implies a 31° counter-clockwise rotation (Table 4) during the Tertiary. The paleomagnetic data of Cretaceous

Table 4
Previously published paleomagnetic results from the southern Apennines and the Apulian foreland and comparison to the results from this study

N	D	I	k	Locality	Age	reference
Apulian Unit						
4 sites	324.4	36.0	480	Gargano peninsula	Cret.	Channell and Tarling (1975)
17 sites	335.0	37.8	31	Gargano peninsula	Late Cret.	Channell (1977)
21 sites	327.7	38.2	56	Gargano peninsula	Late Cret.	VandenBerg (1983)
3 studies	**329.0**	**37.4**	**328**	**Gargano peninsula**	**Cretaceous**	VandenBerg (1983)
5 areas	**359.8**	**56.3**	**491**	**Apulian Unit**	**Early Pleist.**	[I]
Bradano Unit						
6 areas	**359.1**	**56.3**	**218**	**Matera area**	**Early Pleist.**	[I], [II] and this study
Southern Apennines Fold Belt, internal part						
3 sites	285.8	54.1	48	Capri (CL platform)	Late Cret. (Senonian)	Catalano et al. (1976)
5 sites	286.6	44.5	21	Mt.Cerviero (CL platform)	Late Cret.	Manzoni (1975)
113 samples	248.0	30.4	48	Pescopagnano (LN basin)	Cret.	Incoronato (1983)
Southern Apennines Fold Belt, external part						
5 sites	322.4	28.2	120	Campia (Matese Mt.)	Cret.	Channell and Tarling (1975)
1 site	322.2	30.9		Maggiore (AC platform)	Late Cret. (Senonian)	Catalano et al. (1976)
9 sites	326.0	42.0	44	Monte Maiella	Cret.	Jackson (1990)
Southern Apennines Fold Belt (summary)						
3 areas	**271.1**	**44.6**	**16**	**Internal part**	**Cretaceous**	Jackson (1990)
3 areas	**323.4**	**33.7**	**116**	**External part**	**Cretaceous**	Jackson (1990)
13 sites	337.8	56.1	111	Sant'Arcangelo basin	Early Pleist.	Sagnotti (1992)
4 sites	**351.3**	**56.1**	**144**	**Apennines Fold Belt**	**middle–Late Pleist.**	this study
20 sites	**336.0**	**58.4**	**76**	**Apennines Fold Belt**	**Early Pleist.**	this study
8 sites	**320.8**	**52.4**	**159**	**Apennines Fold Belt**	**Late Mio–middle Plio**	this study

Symbols are the same as used in Table 3. CL platform = Campania–Lucana platform; LN basin = Lagonegro basin; AC platform = Abruzzi–Campania platform.

rocks from the southern Apennines (Channell and Tarling, 1975; Catalano et al., 1976; Incoronato, 1983; Jackson, 1990) are quite scattered, but they all imply considerable counter-clockwise rotations (Table 4). The sites from the internal part of the Apenninic belt show a larger rotation (89°) than the ones from the external part (37°). This 37° counter-clockwise rotation of the external part of the southern Apennines is in good agreement with the directions in the Late Miocene–middle Pliocene sediments (39°) studied here. Hence, there is (at present) no evidence for any tectonic rotations in the external part of the southern Apenninic belt system between the Cretaceous and the latest Miocene. The present results also indicate a 15° counter-clockwise rotation during the Pliocene and a further 24° counter-clockwise rotation during the Early Pleistocene.

It has been suggested that the whole area of the southern Apennines behaved more or less as one single coherent block during tectonic rotations in the middle Pleistocene on the basis of the seismicity data (Gasparini et al., 1982) and micro-tectonic analysis (Auroux et al., 1986), which indicate that the direction of the "Quaternary compressive phase" is homogeneous for the whole area (Auroux et al., 1986). This is not supported by our data; for instance, the average rotation of 24° counter-clockwise for the Early Pleistocene shows values ranging between 17 and 31° (Table 3; Fig. 8A) for the individual localities.

The differential rotation in the time between the Late Miocene–middle Pliocene (39°) and the Early Pleistocene (24°) sediments for the southern Apennines Fold Belt is 15° counter-clockwise. This middle–Late Pliocene 15° counter-clockwise rotation for the southern Apennines Fold Belt contrasts with a 10° clockwise rotation for the Caltanissetta basin of Sicily, dated earlier at approximately 3.05 Ma (Scheepers and Langereis, 1993). The subsequent middle Pleistocene phase also shows opposing rotations for the two regions, 24° counter-clockwise for the southern Apennines Fold Belt and 15° clockwise for the Calabro-Peloritan block.

A final note concerns the inclinations in our Pleistocene sediments, which on average are 56–58° (Table 4). These values are close to the expected axial dipole field inclination (59.7) for the present latitude of the area (~ 40.5°N). This implies that the inclination error in these low-carbonate marine clays is small (2–4°). The inclination difference in the Late Miocene–middle Pliocene marls is slightly larger (~ 8°), but this is not caused by a northward movement, and simply reflects a larger inclination error, most likely related to a higher carbonate content (Scheepers and Langereis, 1993). The large inclination difference of the Cretaceous data cannot be explained by an inclination error only and suggests northward movements since the (Late) Cretaceous. A more extensive discussion of this and previous work in the Sicilian–Calabrian arc system and the consequences for the geodynamical history and setting of the central Mediterranean area is given by Van Dijk and Scheepers (1994).

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